

VISHAL VARWANI

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EDUCATION

SRH Hochschule Heidelberg | MSc Applied Data Science and Analytics | Heidelberg, Germany | April 2024 – April 2026

Somaiya Vidyavihar University | Bachelor of Computer Application | Mumbai, India | March 2020 – April 2023

TECHNICAL SKILLS

GenAI & Agents: LangChain, LangGraph, RAG, hybrid search (dense + sparse), cross-encoder reranking, agentic workflows, OpenAI API, Hugging Face, Groq, Mistral, LLM fine-tuning, LoRA, PEFT

ML & Data Science: PyTorch, TensorFlow, scikit-learn, LightGBM, BERT, T5, Llama, NLP, LSTM, Explainable AI (LIME, SHAP), time series forecasting, A/B testing, NDCG, MRR

Engineering: Python, SQL, JavaScript, TypeScript, Flask, FastAPI, React, Node.js, Docker Compose, Kubernetes, GCP (BigQuery, Cloud Run, Cloud Storage), Supabase, Vercel, Render

Data Engineering: Kafka, Spark Structured Streaming, Delta Lake, Snowflake, PostgreSQL, MongoDB, Redis, Airflow, dbt, n8n

Vector Search: Pinecone, pgvector, Chroma DB, FAISS, BigQuery Vector Search, Voyage AI embeddings, Sentence-Transformers, BM25

Evaluation & MLOps: DeepEval, pytest, evaluation dataset design, behavioral criteria testing, model drift detection, JSON schema enforcement

EXPERIENCE

e.Ray Europa GmbH — Working Student – AI Engineer

Darmstadt, Germany | April 2025 – April 2026

- Built a LangGraph multi-agent monitoring system polling 15+ government portals: an ingestion agent fetched new publications, a quality agent validated schema compliance and flagged anomalies, and a routing agent decided whether to auto-process or queue for human review, reducing manual pipeline monitoring time by ~70%.
- Implemented domain RAG with hybrid search combining dense Voyage AI embeddings in Pinecone and sparse BM25 retrieval with cross-encoder reranking, achieving ~88% query accuracy across 1,200+ indexed environmental documents and reducing retrieval from manual hours to seconds.
- Diagnosed a ~35% pipeline failure rate caused by partial extractions, hallucinated regulatory values, and mixed prose in LLM output; applied chain-of-thought prompting and strict JSON schema enforcement, dropping the error rate to under 2%.
- Built Flask APIs handling 2,000+ daily ingestion calls from live sensor streams; applied rate limiting, request queuing, and in-memory entity caching, improving throughput by ~50% with a ~70% cache hit rate.

Pareto — Research Student – Sales Forecasting

Heidelberg, Germany | November 2024 – March 2025

- Developed an hourly demand forecasting model in PyTorch for retail and marketplace clients across high-variance product categories in two country markets, improving forecast accuracy by 12-18% over baseline and reducing overstock and stockout prediction errors by ~40%.
- Designed the end-to-end data pipeline querying raw sales records from PostgreSQL with SQL transformations resolving missing values, schema mismatches, and multi-country format inconsistencies before model training.
- Built a rolling-origin backtesting framework with automated hyperparameter sweeps and structured experiment logs, reducing overconfident spike predictions by ~40% across iterative model runs.

Wipro — Data Engineer Intern

Mumbai, India | April 2023 – January 2024

- Built and maintained Python ETL pipelines ingesting raw transaction data from 5 client REST APIs into a centralized PostgreSQL data warehouse, normalizing schemas across inconsistent source formats and processing 50M+ records monthly.
- Designed Airflow DAGs orchestrating daily ingestion, transformation, and aggregation workflows across 3 retail client pipelines; added SLA breach alerts and failure notifications, cutting undetected pipeline downtime by ~80%.
- Built pipeline observability on GCP using BigQuery and Looker Studio tracking row counts, latency, and error rates per run, enabling the data team to detect upstream feed anomalies within minutes rather than hours.
- Built a REST API data connector pulling live inventory and sales records on 15-minute intervals with exponential backoff retry logic and a dead-letter queue, achieving 99.4% ingestion reliability.

PROJECTS

JobMate | github.com/VishalVarwani/Jobscan | Sep 2024 – Mar 2025

Full-stack job marketplace aggregating 20,000+ listings across 4 boards. Grew to 1,000 organic users with zero paid marketing.

- Implemented a two-stage ranking system: first-stage dense retrieval using Sentence-Transformers filtered candidates from the full listing index; second-stage cross-encoder reranking personalised results to each user's resume profile, achieving ~95% top-1 accuracy on a held-out evaluation set.
- Designed offline A/B experiments comparing BM25, dense retrieval, and cross-encoder strategies using NDCG@10 and MRR; cross-encoder reranking improved NDCG@10 by ~18% over the semantic-only baseline.
- Decoupled LLM inference as an async background job with a polling callback, reducing main API response times from 10-12 seconds to under 400ms.

Invoice-to-Pay Agent | github.com/VishalVarwani/invoice-to-pay | Jan 2026 – Apr 2026

Open-source accounts-payable automation agent covering the full AP workflow from document upload to ERP posting.

- Built a deterministic LangGraph control graph processing invoices through schema validation, duplicate detection, 2-way and 3-way PO matching, fraud controls, GL coding, and ERP posting; a human approval interrupt required explicit sign-off before any ERP write.
- Enforced strict Pydantic schemas across invoices, POs, delivery notes, and audit records; dual-parser routing directed structured documents to LiteParse and complex layouts to Docling.
- Covered the full graph, API layer, and parser contracts with pytest; replaced in-memory checkpointing with Supabase for persistent workflow state and full audit logs.

Supply Chain Monitoring Agent | github.com/VishalVarwani/Supply-Chain-Monitoring | Mar 2026 – May 2026

LangGraph multi-agent system for real-time supply chain anomaly detection and procurement policy validation.

- Built three specialised agents: a data ingestion agent pulling inventory and order records from supplier REST APIs, a validation agent running LLM-powered compliance checks via structured tool calls, and a routing agent escalating anomalies by severity.
- Constructed an evaluation dataset of 300+ labeled supply chain events and ran iterative preference-based improvement loops, reducing false positive escalations by ~25% over three feedback cycles.
- Deployed via Docker Compose with DeepEval behavioral criteria tests running before each release, catching agent regression without exact string matching.

ACHIEVEMENTS

- **Winner – BioMed X Hackathon** (Heidelberg, March 2025): Built a healthcare knowledge-graph agent in 24 hours ingesting biomedical PDFs, running NER and entity linking over gene, disease, and drug mentions. Post-hackathon pilot cut manual literature triage by ~30%. First place.
- **Top 5 – Zurich Datathon 2025, Alpiq Challenge** (April 2025): Built a forecasting pipeline for hourly energy consumption across two inconsistent country portfolios, spending 60% of project time on data quality. Top 5 of 50+ teams.

ACTIVITIES

Project Manager, Google Developer Groups (GDG) Mannheim — March 2025 to present. Co-organised meetups, hack nights, and hackathons. Facilitated ML and cloud sessions for 50+ members and onboarded new contributors.